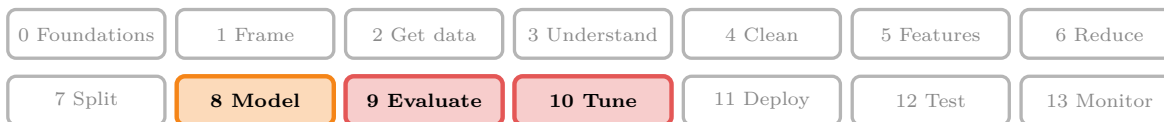


Bagging Regressor

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Where this fits: Pipeline map steps 8 (Model), 9 (Evaluate), 10 (Tune). See the [Course map](#).



- **Builds on:** Simple imputation (mean, median, mode, constant) ([Note 37](#)); ML pipelines ([Note 38](#)); Bias-variance trade-off ([Note 62](#)); Overfitting ([Note 91](#)); Hyperparameter tuning ([Note 98](#)); Decision trees ([Note 100](#)).
- **Leads to:** Random forest ([Note 108](#)).
- **Compare with:** Boosting ([Note 101](#)); Voting ensembles ([Note 104](#)); Cross-validation ([Note 104](#)); Random forest ([Note 108](#)); Optuna ([Note 134](#)); Bayesian optimisation ([Note 134](#)).

1. Overview

Key point: A bagging regressor works exactly like a bagging classifier, except that aggregation takes the mean of the base models' numbers instead of a majority vote.

The bagging regressor applies bagging (the [bagging Note](#)) to regression. Everything carries over from the [bagging classifier Note](#): the four sampling types and every hyperparameter. This Note covers:

- the one difference, in the aggregation step;
- a demo with curves: one tree against a bagged ensemble;
- `BaggingRegressor` on the Boston housing data, tuned with `GridSearchCV`.

The Notebook (`notebook.ipynb`) runs every experiment. The Dash app `app.py` lets us change the base model and the bagging settings and redraws both curves.

2. The core idea

Key point: Bootstrapping is unchanged; aggregation averages the predictions.

In the **bootstrapping** step, each base model is trained on its own random sample of the data, as before. So pasting, random subspaces and random patches are all available too.

In the **aggregation** step, each base model returns a number, since this is regression. The bagging regressor returns their **mean**. That is the only change.

scikit-learn's `BaggingRegressor` has exactly the same hyperparameters as `BaggingClassifier` (the [bagging classifier Note](#), section 6): `estimator`, `n_estimators`, `max_samples`, `max_features`, `bootstrap`, `bootstrap_features` and `oob_score`.

3. Seeing it on a curve

Key point: One fully grown tree passes through every training point (test R^2 0.92); 50 bagged trees give a smooth curve that follows the pattern (0.95).

The demo data has one input column. Its output is two Gaussian bumps (a small one at 0 and a larger one at 2) plus a little noise, with 150 training points and 100 test points. With a single input there are no columns to sample, so the app only offers row settings: the base model (decision tree, SVR or KNN), `n_estimators`, `max_samples` and `bootstrap`.

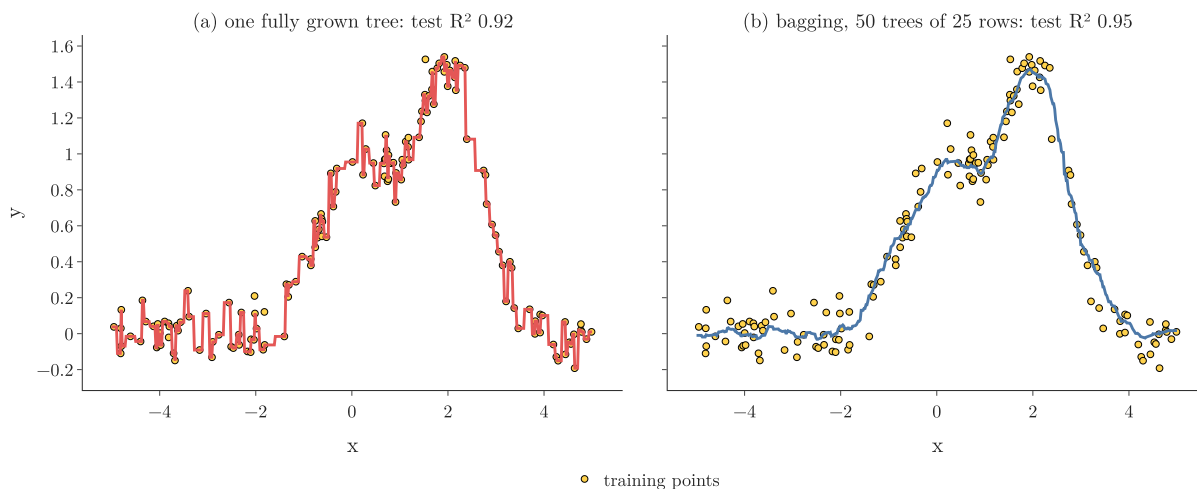


Figure 1: (a) One fully grown regression tree; (b) a bagging regressor of 50 trees, each trained on 25 rows drawn with replacement

Figure 1 compares a single decision tree with bagging (50 trees, 25 rows each, with replacement):

- **(a) The single tree** tries to pass through every training point, outliers included: low bias, high variance, so it overfits. Test R^2 : **0.92**.
- **(b) The bagging regressor** gives a smooth curve. It does not touch every point; it follows the underlying pattern. Test R^2 : **0.95**.

This is the same effect as in classification: bagging keeps the low bias and removes much of the variance.

Changing the settings makes little difference on this easy data: 100 trees score 0.952, and pasting (75 rows, `bootstrap=False`) scores 0.947. Bagging is not always an improvement, though: bagged KNN (0.832) and bagged SVR (0.871) both do worse than a single KNN (0.954) or SVR (0.894) here.

4. BaggingRegressor on the Boston housing data

Key point: Out of the box, bagged trees (test R^2 0.83) beat linear regression (0.66), a single tree (0.38) and KNN (0.55).

4.1 Three single regressors

Key point: Linear regression 0.66, decision tree 0.38, KNN 0.55 on one test split.

The Boston housing data (the [regression trees Note](#), section 7.1) has 506 districts, 13 input columns and the median house price. We split it into 404 training rows and 102 test rows, and train three single regressors:

Model	Linear regression	Decision tree	KNN
Test R^2	0.659	0.380	0.548

4.2 A bagging regressor with default settings

Key point: Ten bagged trees, nothing tuned: training R^2 0.98, test R^2 0.83.

Python: A bagging regressor with every setting at its default.

```
from sklearn.ensemble import BaggingRegressor
```

```
bag = BaggingRegressor(random_state=1)
bag.fit(X_train, y_train)
bag.score(X_train, y_train)    # 0.980
bag.score(X_test, y_test)     # 0.828
```

The defaults: `estimator=None` (a decision tree), `n_estimators=10`, every training row per model (`max_samples=None`), drawn with replacement, and every column. `score` returns R^2 for a regressor.

Without tuning anything, the bagging regressor beats all three single models. The gap between training (0.98) and test (0.83) shows some overfitting remains.

4.3 Tuning with GridSearchCV

Key point: A grid search over 144 combinations picks 50 bagged trees on full bootstrap samples; cross-validation R^2 0.87, test R^2 0.81.

Instead of trying pasting, random subspaces and random patches by hand, we let `GridSearchCV` try them all (the [KNN Note](#), section 4.2):

Python: Searching over the base model and the sampling settings.

```
params = {
    "estimator": [None, LinearRegression()],
```

```

        KNeighborsRegressor()],
    "n_estimators": [20, 50, 100],
    "max_samples": [0.5, 1.0],
    "max_features": [0.5, 1.0],
    "bootstrap": [True, False],
    "bootstrap_features": [True, False]}
grid = GridSearchCV(BaggingRegressor(random_state=1),
                    param_grid=params, cv=3,
                    n_jobs=-1, verbose=1)
grid.fit(X_train, y_train)
grid.best_params_

```

The base model itself is a hyperparameter here: `None` stands for the default decision tree. With `bootstrap` and `bootstrap_features` both tried as `True` and `False`, the search covers bagging, pasting, random subspaces and random patches.

That is $3 \times 3 \times 2 \times 2 \times 2 \times 2 = 144$ combinations, each with 3-fold cross-validation: 432 fits, 13 seconds with `n_jobs=-1`. The best settings:

- base model: **decision tree**;
- `bootstrap=True`: **bagging**, not pasting;
- `bootstrap_features=False`, `max_features=1.0`: **no column sampling**;
- `max_samples=1.0`: every model gets as many rows as the training set;
- `n_estimators=50`.

Their cross-validation R^2 is **0.871**, and the test R^2 is **0.806**.

Extra: The 0.871 is an average over the three validation folds of the training data; the 0.806 is measured on the 102 test rows. The two differ because they are measured on different rows, and 102 rows is a small, noisy test. Expecting the test score to match `best_score_` exactly is a common mistake. Here the tuned model scores slightly lower on this test set than the default one (0.828); the search picks settings that do best on average, not on this particular test set.

4.4 The out-of-bag score

Key point: With `oob_score=True`, the regressor reports R^2 on the out-of-bag rows: 0.870 here.

`oob_score` works for regression too (the [bagging classifier Note](#), section 4). For a regressor, `oob_score_` is the R^2 on the rows each model never saw. With 50 trees it is **0.870**, close to the cross-validation score.

5. Summary

	Bagging classifier	Bagging regressor
Base models	classifiers	regressors
Aggregation	majority vote	mean
scikit-learn	<code>BaggingClassifier</code>	<code>BaggingRegressor</code>
score and <code>oob_score_</code>	accuracy	R^2
Hyperparameters	<code>estimator</code> , <code>n_estimators</code> , <code>max_samples</code> , <code>max_features</code> , <code>bootstrap</code> , <code>bootstrap_features</code> , <code>oob_score</code>	the same

- Only aggregation changes for regression: the mean replaces the vote.
- On the bumps data, bagging smooths one overfitting tree into a curve that follows the pattern: test R^2 0.92 to 0.95.
- On Boston, default bagged trees score 0.83 against 0.66 for the best single model; a grid search picks plain bagging of 50 trees (cross-validation 0.87).

6. Key terms

Term	Meaning
Bagging regressor	A bagging ensemble of regressors that predicts the mean of their predictions
<code>BaggingRegressor</code>	scikit-learn class for bagging, pasting, random subspaces and random patches in regression